

Nonlinear relationships and transformations

Last time: Class activity

$$\widehat{\text{Life Expectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

What is the predicted ^{life expectancy} ~~price~~ for GDP per capita of \$20200?

$$4.95 + 7.20 \log(20200) = 76.327$$

20000 $\rightarrow$ 20200 is 1% increase

Class activity

$$\widehat{\text{LifeExpectancy}} = 4.95 + 7.20 \log(\text{GDP per capita})$$

+ When GDP per capita = 20000: $\widehat{\text{LifeExpectancy}} = 76.255$

+ When GDP per capita = 20200: $\widehat{\text{LifeExpectancy}} = 76.327$

Change:

$$\begin{aligned} 76.327 - 76.255 &= 0.072 \\ &= \frac{7.20}{100} \end{aligned}$$

Interpretation : a 1% increase in GDP per capita is associated with a change of 0.072 years in life expectancy, on average

Class activity

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

+ Price in \$, Area in hundreds of sq ft

What is the predicted price for a house of 2100 square feet?

$$\widehat{\text{Area}} = 21$$

$$\widehat{\log(\text{Price})} = 10.51 + 0.091(21)$$

$$= 12.421$$

$$\Rightarrow \widehat{\text{Price}} = e^{12.421} = \$ 247954$$

Class activity

20 $\rightarrow$ 21 (one unit increase in Area)
(an increase of 100 sq ft)

$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

- + Price in \$, Area in hundreds of sq ft
- + When Area = 20: $\widehat{\text{Price}} = 226386$
- + When Area = 21: $\widehat{\text{Price}} = 247954$

Change:

$$\frac{247954}{226386} = 1.095 = e^{0.091}$$

Class activity

$$e^x \approx 1 + x$$

x close to 0

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

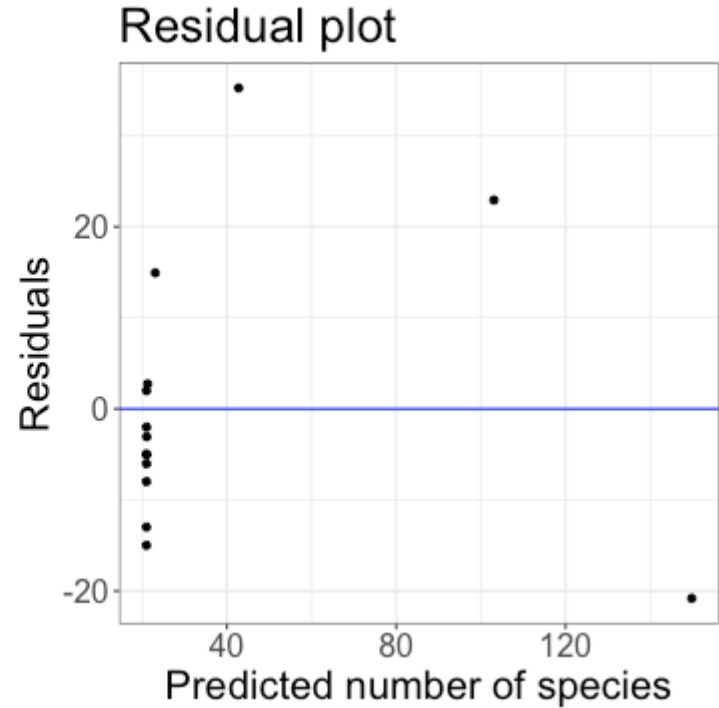
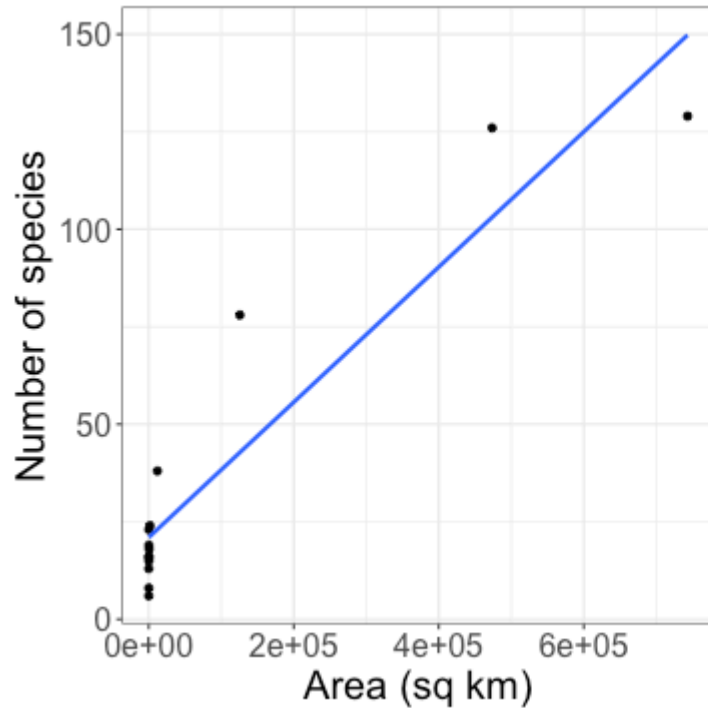
$$\widehat{\log(\text{Price})} = 10.51 + 0.091 \text{ Area}$$

- + Price in \$, Area in hundreds of sq ft
- + When Area = 20: $\widehat{\text{Price}} = 226386$
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$$\text{Change: } \frac{247954}{226386} = 1.095 = e^{0.091}$$

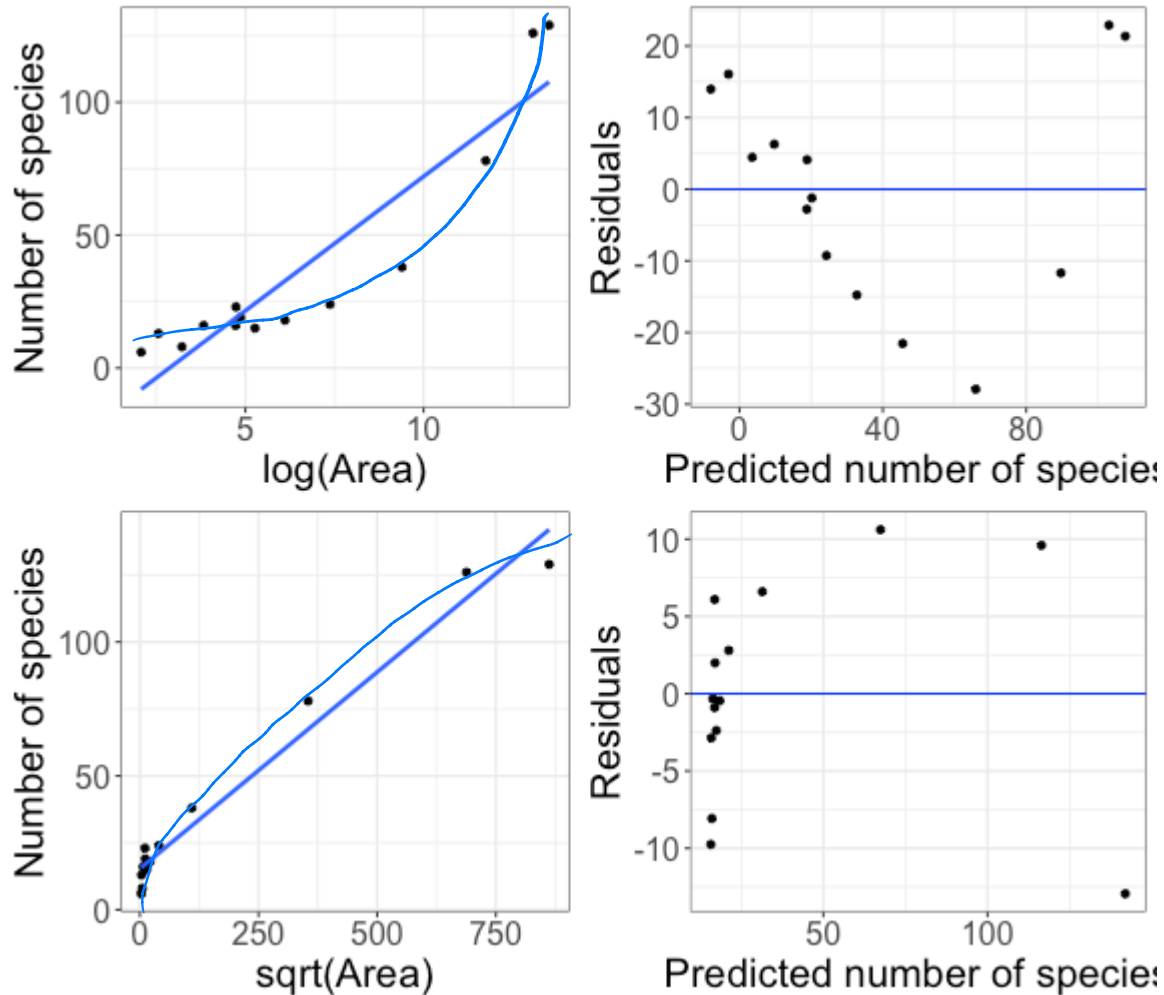
Interpretation : a one-unit increase in Area is associated with an increase in price by a factor of $e^{0.091} = 1.095$, on average

More transformations



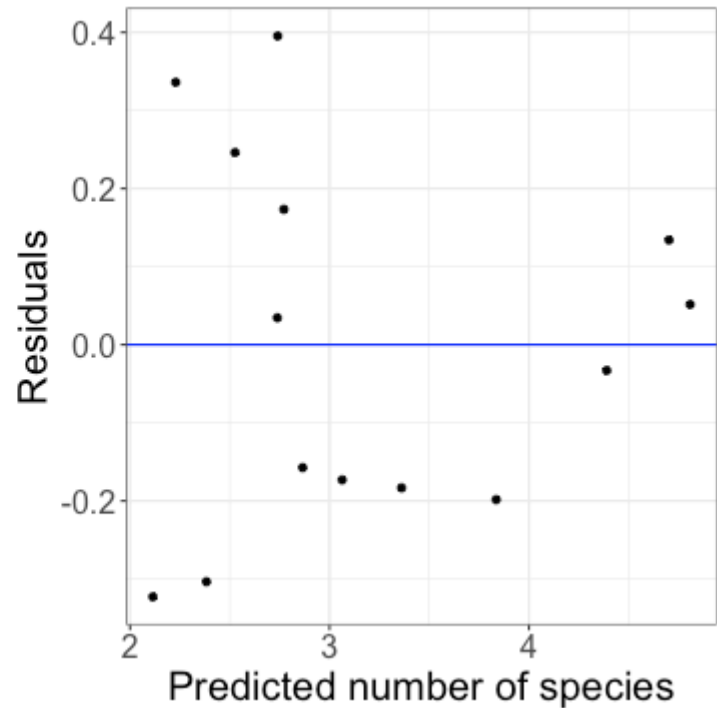
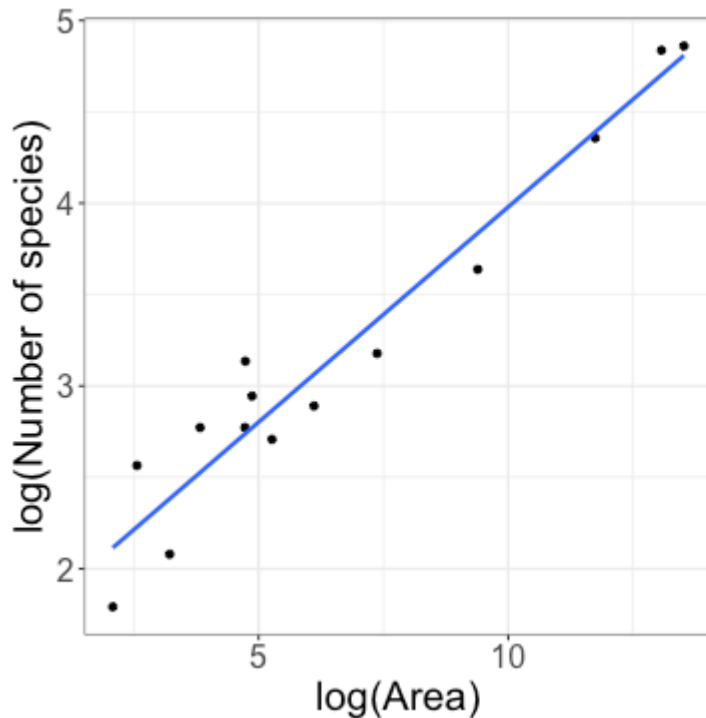
What transformations could we try here?

Trying some transformations



Transform both predictor and response?

$$\log(\textit{Species}) = \beta_0 + \beta_1 \log(\textit{Area}) + \varepsilon$$



Interpreting the fitted model

$$\log(\widehat{Species}) = 1.625 + 0.246 \log(Area)$$

$$Area = x$$

$$\log(\widehat{Species}) = 1.625 + 0.246 \log(x)$$

$$\begin{aligned} \widehat{Species} &= e^{1.625 + 0.246 \log(x)} \\ &= e^{1.625} e^{0.246 \log(x)} \\ &= e^{1.625} e^{\log(x^{0.246})} \\ &= e^{1.625} (x)^{0.246} \end{aligned}$$

$$\frac{e^{1.625} (1.01x)^{0.246}}{e^{1.625} (x)^{0.246}} = \frac{(1.01)^{0.246} \cancel{x^{0.246}}}{\cancel{x^{0.246}}}$$

$$= \frac{(1.01)^{0.246} \cancel{x^{0.246}}}{\cancel{x^{0.246}}} = (1.01)^{0.246}$$

$$= (1.01)^{0.246} = 1.0025$$

$$e^{a+b} = e^a e^b$$

$$a \log(b) = \log(b^a)$$

$$e^{\log(a)} = a$$

$$Area = 1.01x$$

$$1.625 + 0.246 \log(1.01x)$$

$$1.625 + 0.246 \log(1.01x)$$

$$e^{1.625} e^{0.246 \log(1.01x)}$$

$$e^{1.625} e^{\log((1.01x)^{0.246})}$$

$$e^{1.625} (1.01x)^{0.246}$$

Interpretation:

a 1% increase in Area is associated with
an increase in # of species by a factor of
 $(1.01)^{0.246}$, on average

Summary of model interpretation so far

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

- + A one-unit increase in x is associated with a change in y of $\hat{\beta}_1$, on average

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \log(x)$$

- + A 1% increase in x is associated with a change in y of $\hat{\beta}_1/100$, on average

$$\widehat{\log(y)} = \hat{\beta}_0 + \hat{\beta}_1 x$$

- + A one-unit increase in x is associated with a change in y by a **factor** of $e^{\hat{\beta}_1}$, on average

$$\widehat{\log(y)} = \hat{\beta}_0 + \hat{\beta}_1 \log(x)$$

$$(1.01)^{\hat{\beta}_1}$$

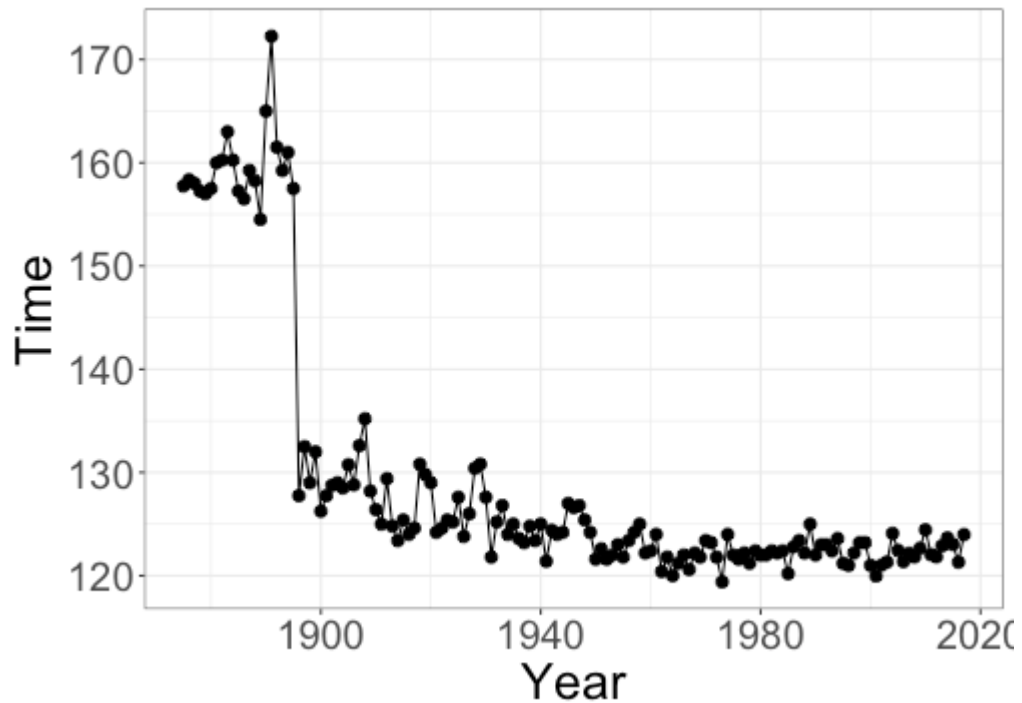
- + A 1% increase in x is associated with a change in y by a **factor** of $e^{\hat{\beta}_1/100}$, on average

Kentucky Derby data

Data on the Kentucky Derby winner from each year between 1875 and 2012. Variables include

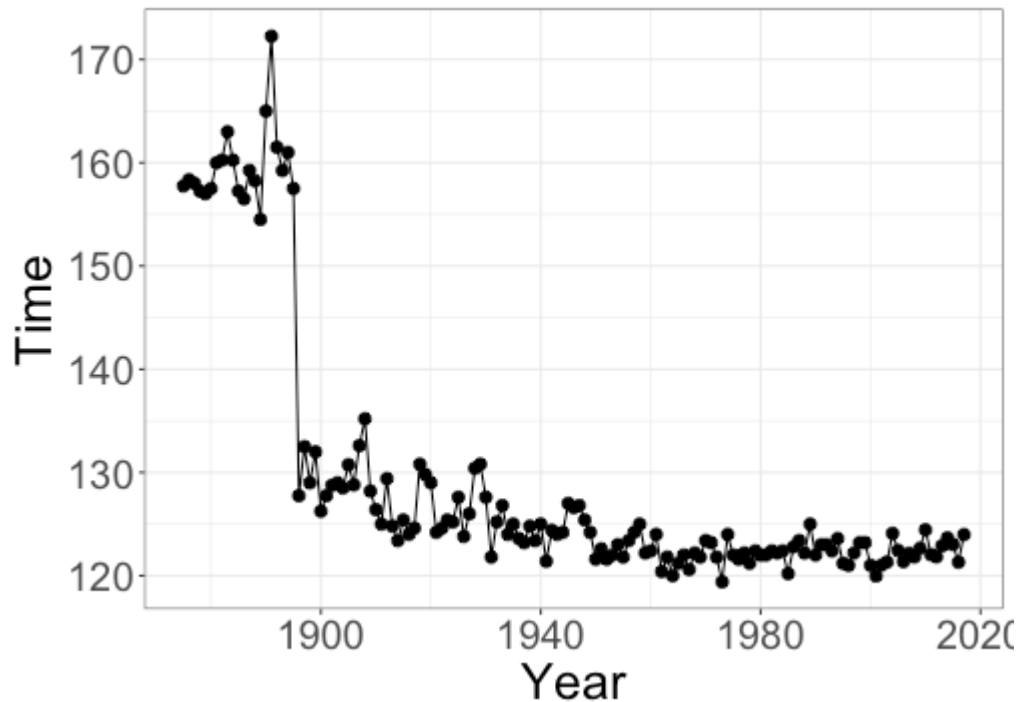
- + year: year of race
- + speed: speed of winning horse (mph)
- + time: winning time (seconds)
- + condition: condition of the track on the day of the race

EDA



What do you notice about the winning times?

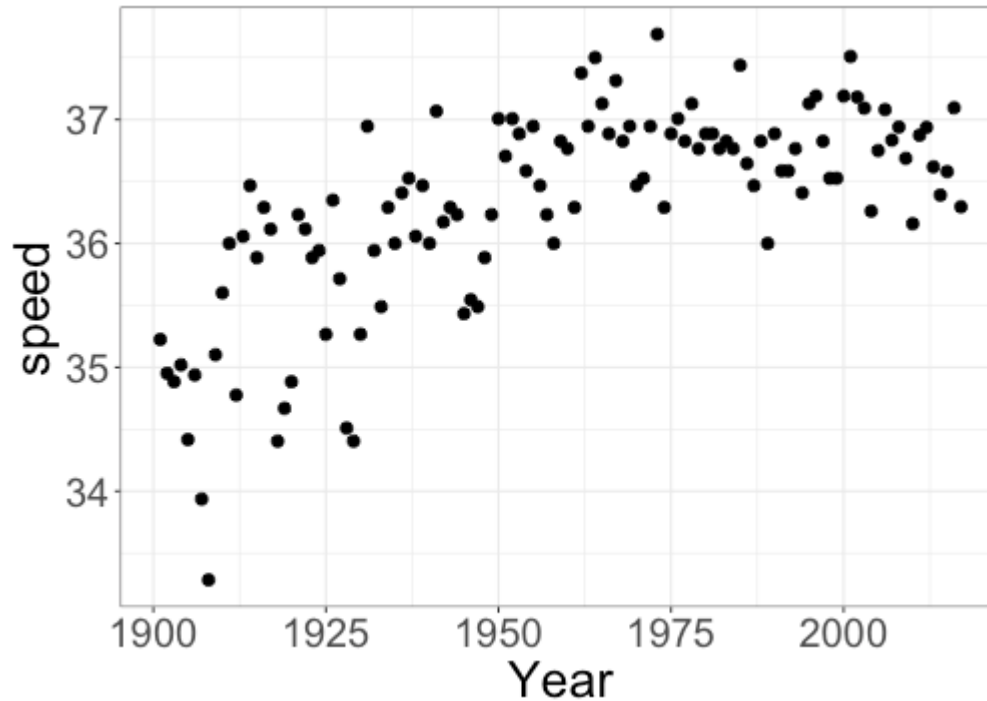
EDA



What do you notice about the winning times?

There's a big drop around 1900, then they decrease gradually.

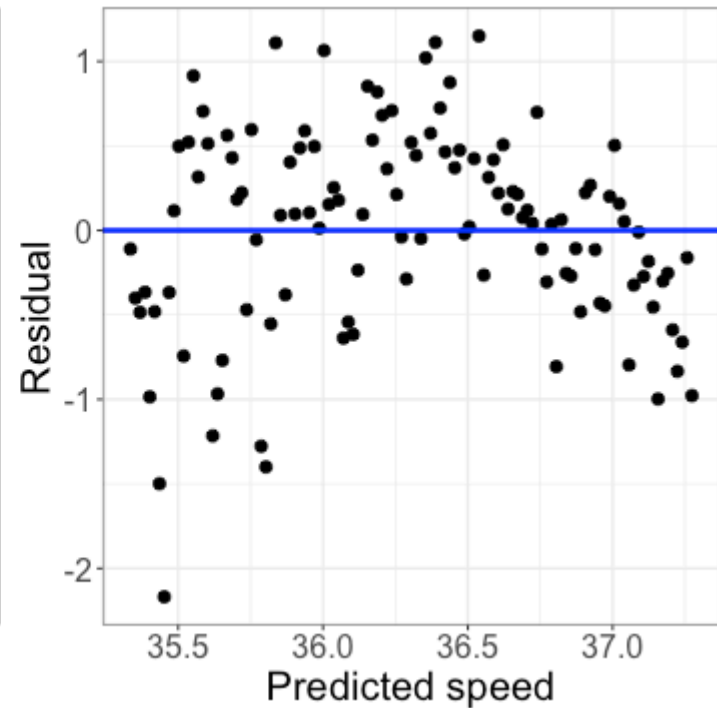
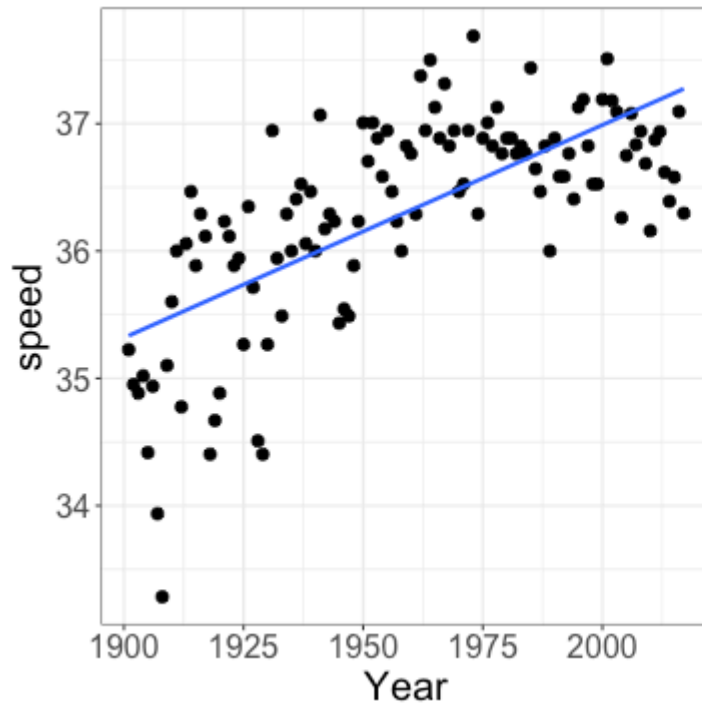
EDA



How would you describe the relationship?

Fitting a linear regression

$$\text{Speed} = \beta_0 + \beta_1 \text{Year} + \varepsilon$$



Polynomial transformations

Another possible transformation is a **polynomial**.

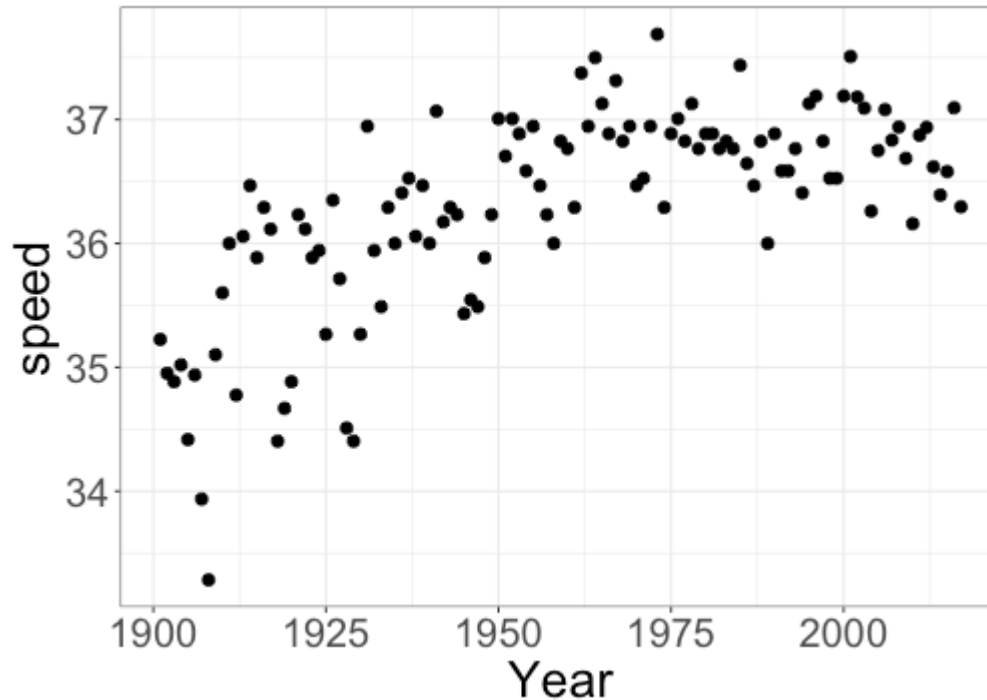
Definition: A polynomial transformation of order k is a linear combination of the terms $X, X^2, \dots, X^k$:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_k X^k + \varepsilon$$

+ Example: If we use a polynomial of order 3, our model is

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$$

Polynomial transformations

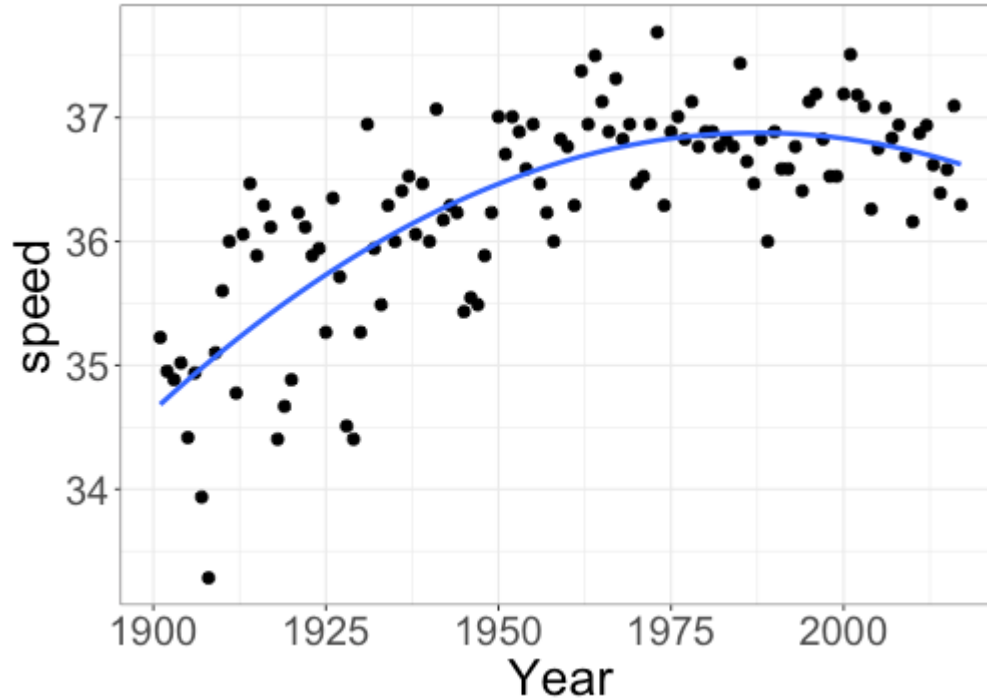


What order polynomial do you think we should use?

Polynomial transformation

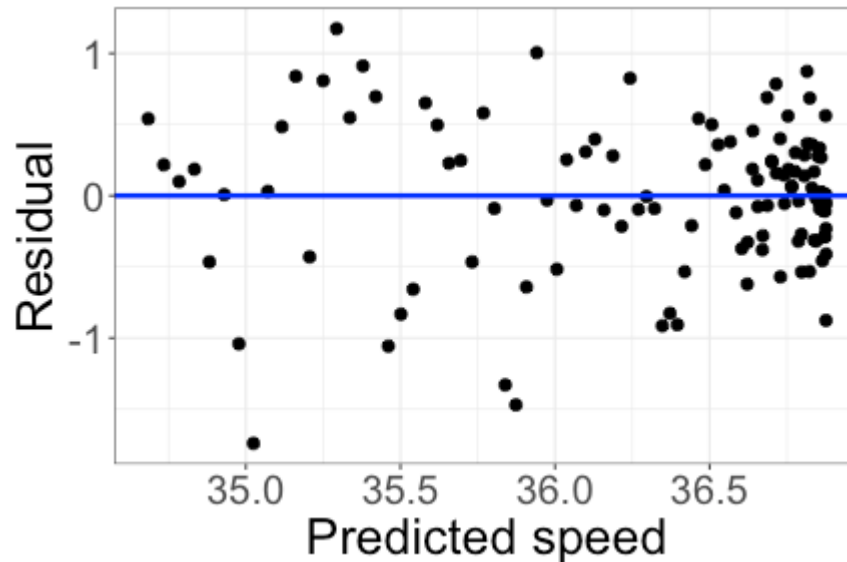
Let's try a polynomial of order 2:

$$\text{Speed} = \beta_0 + \beta_1 \text{ Year} + \beta_2 \text{ Year}^2 + \varepsilon$$



Diagnostic plots

We use residual plots to check the model assumptions after transforming our predictor, just like with other transformations.



How does the shape assumption look now?

Fitting the model in R

```
derby_lm <- lm(speed ~ poly(Year, 2, raw=TRUE),  
               data = derby)
```

- + `poly` means "fit a polynomial"
- + `poly(..., 2, ...)` specifies a second-order polynomial (quadratic)

Fitting the model in R

```
derby_lm <- lm(speed ~ poly(Year, 2, raw=TRUE),  
               data = derby)  
derby_lm
```

```
##  
## Call:  
## lm(formula = speed ~ poly(Year, 2, raw = TRUE), data = d  
##  
## Coefficients:  
##              (Intercept)  poly(Year, 2, raw = TRUE)1  
##              -1.120e+03                1.164e+00  
## poly(Year, 2, raw = TRUE)2  
##              -2.928e-04
```

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

How would I interpret the intercept of the fitted polynomial regression model?

Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

How would I interpret the intercept of the fitted polynomial regression model?

If Year = 0, we predict a speed of -1120 mph (doesn't make much sense, since year is nowhere near 0)

Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

How would I interpret the terms on Year and Year²?

Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

How would I interpret the terms on Year and Year²?

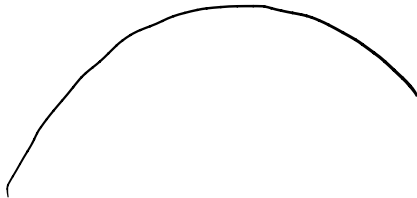
They're kind of difficult to interpret -- if I change Year, I also change Year²

Interpreting the fitted model

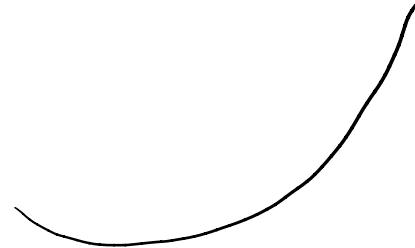
$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

Can I interpret the *sign* on Year^2 ?

negative



positive



Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

Can I interpret the *sign* on Year^2 ?

Yes (for a polynomial of order 2 specifically). The sign on Year^2 tells us about the concavity of the relationship.

Class activity

https://sta112-f25.github.io/class_activities/ca_11.html

- + Experiment with polynomial regression
- + ~~Submit HTML at end of class~~